

1. List all the main components you find inside a desktop PC after opening the cabinet, and write a short description (1-2 lines) for each.

Component	Short Description
Motherboard	The main circuit board that connects and allows communication between all computer components.
CPU (Processor)	Executes instructions and performs calculations; often called the brain of the computer.
CPU Cooler (Fan + Heatsink)	Removes heat from the processor to maintain safe operating temperatures.
RAM (Memory)	Temporarily stores data and programs that are currently being used by the computer.
Storage Drive (HDD/SSD/NVMe SSD)	Permanently stores the operating system, applications, and user files.
Graphics Card (GPU)	Processes images, videos, and graphics for gaming and multimedia applications.
Power Supply Unit (PSU)	Converts AC power from the wall into DC power required by the computer components.
Case Fans	Improve airflow inside the cabinet and help cool internal components.
Heat Sink	Absorbs and dissipates heat from components such as the CPU and GPU.
Optical Drive (DVD/CD Drive) <i>(if installed)</i>	Reads and writes CDs, DVDs, and other optical discs.
Expansion Cards	Add extra features such as Wi-Fi, sound, or additional ports.

Component	Short Description
CMOS Battery	Powers the BIOS settings and system clock when the computer is turned off.
SATA/Data Cables	Transfer data between storage devices and the motherboard.
Power Cables	Deliver electrical power from the PSU to various components.
Computer Case (Cabinet)	Houses and protects all internal components while providing airflow.

2. Take clear photos or draw labeled diagrams of each step as you disassemble a desktop PC, and organize them in a document with captions explaining what you did at each step.

Hint: Include steps like unplugging power, removing side panel, disconnecting cables, and removing RAM, HDD/SSD, and motherboard.

3.Reassemble the PC you disassembled, documenting each step with a brief explanation of what you connected or installed and why the order matters.

4.Create a checklist of safety precautions and tools needed before starting the PC assembly/disassembly process and explain why each is important.

Item	Type	Why It Is Important
✓ Turn Off the PC and Unplug Power Cable	Safety Precaution	Prevents electric shock and protects components from damage.
✓ Press the Power Button After Unplugging	Safety Precaution	Discharge any remaining electrical energy inside the system.
✓ Wear an Anti-Static Wrist Strap	Safety Precaution	Prevents electrostatic discharge (ESD), which can damage sensitive components.
✓ Work on a Clean, Non-Carpeted Surface	Safety Precaution	Reduces dust and minimizes static electricity buildup.
✓ Handle Components by Their Edges	Safety Precaution	Prevents damage to chips, connectors, and circuits.
✓ Keep Liquids Away from the Work Area	Safety Precaution	Avoids short circuits and accidental damage.
✓ Organize and Label Screws	Safety Precaution	Makes reassembly easier and prevents losing small parts.
🔧 Phillips Screwdriver	Tool	Used to remove and install screws in the cabinet and components.
🔧 Anti-Static Wrist Strap	Tool	Protects internal components from static electricity.
🔧 Small Container or Magnetic Tray	Tool	Keeps screws and small parts organized.
🔧 Flashlight	Tool	Helps see connectors and components inside the cabinet.

Item	Type	Why It Is Important
 Compressed Air Can	Tool	Removes dust from fans, heatsinks, and other components.
 Cable Ties	Tool	Keeps cables neat and improves airflow inside the PC.
 Thermal Paste (if replacing CPU cooler)	Tool	Ensures efficient heat transfer between the CPU and heatsink.
 Soft Brush or Microfiber Cloth	Tool	Used to safely clean dust from components.